Mantasha Altab Noyela

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EDUCATION

PhD in Geospatial Analytics | North Carolina State University, Raleigh **Aug 2026 – Present**
MSc in Computer Science | Portland State University, Portland **Sept. 2024 – June 2026**
CGPA: 3.85
BSc in Computer Science & Engineering | Daffodil International University, Bangladesh **May 2016 – Nov. 2020**
CGPA: 3.48

TECHNICAL SKILLS

Languages: C, C++, C#, Python, Java
Core Engineering: Data Structures & Algorithms, OOP, Design Patterns, Debugging, Memory & Performance Optimization, Git
Systems & Backend: Data Pipelines, CSV/XML Processing, Data Validation, Statistical Computing, Visualization
AI / ML: Machine Learning, Deep Learning, Optimization Algorithms, Computer Vision
Databases: MySQL, Microsoft SQL Server
Tools & Frameworks: DevExpress, MuJoCo, YOLOv4
Platforms & Practices: Linux, Windows, Agile, SDLC, Unit Testing

EXPERIENCE

Intel **June 2025 – Sept. 2025**
Graduate Intern

- Developed an interactive data visualization tool in C# with DevExpress, integrating line graphs and heatmaps to analyze test metrics.
- Implemented a contour plot generator from CSV data, applying mathematical computations and interpolation to produce high-quality visual maps.
- Designed a C# XML parser to extract coordinate data and compute statistical insights (min, max, averages).
- Evaluated Intel's internal large language model (LLM) platform, examining its architecture, prompt-generation pipeline, and hybrid cloud-on-premises design to understand strengths, limitations, and enterprise-scale integration.

Agile and Adaptive Robotics Lab **Sept. 2024 – June 2026**
Graduate Research Assistant

- Developed MuJoCo-based neuromechanical locomotion model with physics-driven motion and sensory feedback.
- Implemented Particle Swarm Optimization (PSO) to tune neural-control parameters, improving gait accuracy by ~25%.
- Integrated SNS framework for adaptive neural control and automated convergence toward optimal locomotion speed.
- Applied AI-based control and optimization for biomechanical systems ensuring reproducible, stable simulations.

A Reaction Engineering and Catalysis Science Laboratory (RECSL) **Sept.2023 – May 2024**
Research Assistant

- Supported statistical process control (SPC), continuous improvement efforts and the management and dissemination of metrology and defect data.
- Generated Defect pareto charts and classifying defects using Yield manager to facilitate collaborative defect reduction efforts.
- Assisted with new tool qualifications and evaluated tool performance.
- Analyzed data from various thin film tools using statistical techniques to minimize process variation.

Automatic Control Lab **Sept.2022 – Nov. 2022**
Research Assistant

- Developed an uncrewed aircraft system (UAS) - based wildfire monitoring system using olfactory and visual sensors for real-time detection.
- Designed a fuzzy inference system to improve smoke detection accuracy during prescribed burns.
- Implemented YOLOv4 deep learning model for efficient smoke identification from images.

List of Publications

- Wang, Lingxiao., Pang, Shuo., Noyela, Mantasha., Adkins, Kevin., Sun, Lulu., and Marwa, El-Sayed., "Vision and Olfactory-Based Wildfire Monitoring with Uncrewed Aircraft Systems," 20th International Conference on Ubiquitous Robots (UR), June 25-28, 2023. DOI: 10.1109/UR57808.2023.10202419
- Shuvo, Sabbir Alom., Noyela, Mantasha Altab., and Hossain, Dr. Md. Fokhray., "To Design and Develop Mobile–Based Birth Registration System for New-Born Baby in Bangladesh," ISSN-2347-4890, vol. 9, no. 6, 2021.
  DOI: 10.26821/IJSHRE.9.6.2021.9608